Supplementary material for

The evolution of lithium isotope signatures in fluids draining actively weathering hillslopes

Earth & Planetary Science Letters

Jon K. Golla[a]*, Marie L. Kuessner[b,c], Michael J. Henehan[c], Julien Bouchez[b], Daniella M. Rempe[d], Jennifer L. Druhan[a,b]

[a]Department of Geology, University of Illinois at Urbana-Champaign, Urbana, Illinois 61801, USA
[b]Institut de Physique du Globe de Paris, Sorbonne Paris Cité, Université Paris Diderot, CNRS, 75238 Paris, France
[c]Section 3.3 Earth Surface Geochemistry, GFZ German Research Centre for Geosciences, Telegrafenberg, 14473 Potsdam, Germany
[d]Jackson School of Geosciences, University of Texas at Austin, Austin, Texas 78712, USA
*Corresponding author. E-mail: jgolla2@illinois.edu

This document contains Supplementary Texts S1-S5 and Figures S1-S9.

S1 Development of reactive transport framework

S1.1 Boundary and initial conditions

The vadose zone is discretized into a 1-D column with a Dirichlet boundary condition at the inlet and a flux boundary condition at the outlet. The δ^7Li value of the infiltrating water at the top of the vadose zone is set to the modern average composition of seawater (δ^7Li = +31‰; Millot et al., 2004; Misra and Froelich, 2012). This boundary condition assumes that the δ^7Li in rainwater in this pristine study area, which is located <15 km from the coast of the Pacific Ocean, is not influenced by anthropogenic inputs that would alter a primarily marine signature (Millot et al., 2010). All Li-bearing minerals are initially given the measured bulk bedrock isotope composition (δ^7Li = -0.57‰), except for the necessary adjustment to a much lighter δ^7Li value (-12‰) of dissolving clays in the upper portion of the domain (subsection 4.2). The model is run on the timescale of over a period of 6 years, or approximately five complete pore volumes of fluid rinsing through the domain to remove any effects associated with the choice of initial condition in the fluid phase. However, we do not refer to this as true "steady state" in that the much longer timescale of solid-phase uplift, weathering, and erosion is not rigorously tested for stability. The intent is to model the evolution of aqueous Li signatures as infiltrating fluids pass through present-day regolith

into the vadose zone validated against the novel fluid samples provided by the Vadose Zone Monitoring System (VMS).

S1.2 Saturated zone batch-reactor model

The difference in the intensity of silicate weathering in the vadose zone versus that in the saturated zone is linked to the disparate timescales over which fluid is stored and routed through each subdomain of the near-surface environment. We approach this difference in reaction timescale by simply extracting the resulting fluid and initial solid-phase compositions from the base of the domain of the 1-D model, and allow this system to continue evolving through time without any explicit treatment of down-slope transport. Although such a batch-reactor model is set up to proceed from conditions derived from the bottom of the reactive transport model, this is not intended to imply that the groundwater Li signatures are directly evolving from the bottommost vadose zone Li compositions, nor that flow or transport pathways have ceased. Rather, the purpose is to consider the trajectory of continued solute evolution and explore whether the characteristic water-rock interactions influencing the vadose zone extend deeper into the hillslope.

S1.3 Incorporation of Li into mineral assemblage

This isotope-enabled model update is based on the minerals determined to dissolve and precipitate as a result of water-rock interactions constrained by major cation chemistry of the VMS fluid samples (Wang, 2019). Smectite, illite, and kaolinite are chosen because they contain compatible cations with similar ionic radius to that of Li^{+}, such as Mg^{2+} or Al^{+3}, which have been found to facilitate Li substituion (Horstman, 1957; Odom et al., 1984; Starkey, 1982). In contrast, Li is omitted from quartz and albite due to previous observations of relatively low concentrations in sedimentary formations (Ronov et al., 1970), as well as from carbonates, which are effectively absent above ∼15 m in the weathering profile (Gu et al., 2020). We set the mole fraction of Li in smectite and illite ($X_{\text{Li-Mg}}$) to 0.01024, which is consistent with the bulk bedrock elemental Li/Mg ratio (Table 1). Li in kaolinite is accomodated by a direct Al^{3+} incorporation ($X_{\text{Li-Al}} = 0.00015$). Although higher partitioning could occur, this incorporation is limited by the low initial Al concentrations in the domain.

S1.4 Treatment of kinetic isotopic fractionation

No isotopic re-equilibration is implemented such that the fractionation factors used in the models are constant throughout space and time. We impose this simplification due to the fact

that Li is incorporated via trace substitution in the selected clays, and thus a disconnection exists between the proximity of a given solid phase to equilibrium with the contemporaneous solute concentrations and the proximity of δ^7Li to isotopic equilibrium between phases. While isotopic re-equilibration has been widely observed in major ions during secondary mineral formation (e.g., Fernandez et al., 2019; Frings et al., 2016), we contend that the capacity of a trace element isotope ratio to re-equilibrate is relatively low in these systems.

The model does not distinguish among the possible mechanisms that underlie this isotopic fractionation (Hindshaw et al., 2019), namely, structural substitution (Vigier et al., 2008; Williams and Hervig, 2005), adsorption (Pistiner and Henderson, 2003; Zhang et al., 1998), and exchange in loosely-bounded sites (Wimpenny et al., 2015). The extent of sorption-derived fractionation is a function of the maturity of the secondary mineral that Li is sorbing onto, suggesting (oxy)hydroxides, like gibbsite produce the largest sorptive fractionations by as much as +30‰ (Millot and Girard, 2007). The solid-phase composition (Gu et al., 2020) used as part of the initial conditions in this model is devoid of these metal hydroxides, and thus we proceed in the use of a single fractionating pathway encompassing the aforementioned potential Li uptake processes in association to the formation of secondary clays. This approach is justified for each of the Li-bearing phases below.

One of the earliest laboratory-isolated Li isotope fractionation factor for kaolinite was from Zhang et al. (1998), who reported an $\alpha_{\text{kaol-diss}}$ for Li uptake via adsorption. Since this study, there have been few attempts to experimentally quantify Li isotope fractionation between kaolinite and the dissolved load during adsorption (Li and Liu, 2020; Zhang et al., 2021) and none for fractionation during kaolinite formation. Of these adsorption-based fractionation factors, Zhang et al. (1998) ($\alpha_{\text{kaol-diss}} = 0.979$) and Zhang et al. (2021) ($\alpha_{\text{kaol-diss}} = 0.976$) arrive at comparable values while the magnitude of fractionation observed by Li and Liu (2020) is significantly lower ($\alpha_{\text{kaol-diss}} = 0.992$). This discrepancy may be a function of the ionic strength of the solution used in the batch experiments. It has recently been demonstrated that Li isotope fractionation becomes larger with increasing extent of adsorption of Li onto kaolinite, which is positively correlated to the solution ionic strength (Li and Liu, 2020). The 0.001-M and 0.1-M solutions used by Li and Liu are more typical of the natural waters found in our study area versus the more saline 0.7-M synthetic seawater solution and Gulf of Mexico water used in the Zhang et al. (2021) and Zhang et al. (1998) experiments, respectively. For this reason, we use the $\alpha_{\text{kaol-diss}}$ from Li and Liu. Furthermore, this value is representative of kinetic fractionation that predominates when pH > 5, which is characteristic of the conditions in our study area. Adsorption of Li onto kaolinite through this mode of fractionation is driven by removal from solution as inner-sphere surface complexes (Li and Liu, 2020). We assume that the strong covalent bonding associated with such

chemisorption will essentially remove the adsorbed Li from the exchangeable pool. Thus, we approximate the isotopic fractionation that results from this process as the fractionation that would be observed during neofromation of kaolinite in the reactive transport model.

We argue that a comparable representative fractionation mechanism (i.e., via secondary mineral formation) is also valid for illite. A previous investigation has demonstrated that there is little to no fractionation during adsorption of Li onto illite (Millot and Girard, 2007). The fractionation factor used for smectite is already weighed for fractionation during precipitation and adsorption (Hindshaw et al., 2019). Recent analysis of soil samples developed from basaltic bedrock seem to indicate the importance of exchangeable Li relative to structural incorporation (Pogge von Strandmann et al., 2021). Although this finding is relevant to soils, we are focused on a weathered bedrock composed of relatively fewer surface exchange sites and lack the appropriate parameterization to explore the alternative scenario for such conditions. Future analysis could incorporate an exchangeable pool should the appropriate exchange coefficients and fractionation factors become available for an actively weathering shale.

S1.5 Model calibration

The vadose zone reactive transport model employed in this work is based on an investigation by Wang (2019), who developed and calibrated the model with major cation solute concentrations, mineralogy, and a series of batch-reactor experiments utilized to constrain water-rock reaction rate laws. The incorporation of Li in this update is accompanied by additional calibration in order to reproduce the overall behavior of both Li trace element concentration and its isotopes (Figure S2). However, it is critical to recognize that none of these changes alter the agreement between simulated behavior and measured major cations reported by Wang (2019) as shown in Figure S3. The fit of these simulations is outside the scope of this study and is discussed in detail in Wang (2019).

S2 Atmospheric correction for well and stream samples

The impact of direct atmospheric inputs and, particularly, the geochemical composition of precipitation in a location proximal to the Pacific Ocean may require corrections. While such impact of rain input is also expected for waters collected in the vadose zone, this correction is deemed unnecessary in the present study as the reactive transport approach allows for a direct comparison between measured and modeled concentrations (subsections 3.3 and 4.2). Therefore, we account for atmospheric inputs only in the well and stream samples by applying a correction for marine aerosols following Dellinger et al. (2015). We assume that

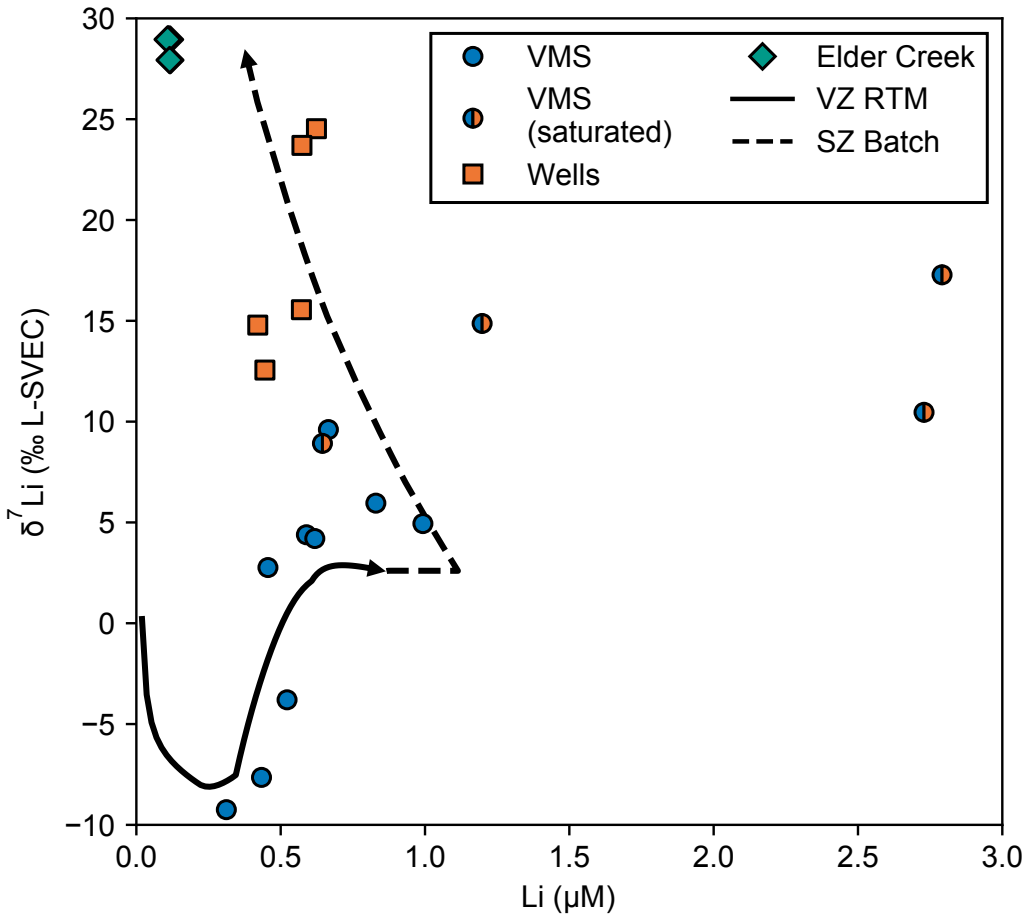

Figure S1: Li concentrations and isotope signatures of VMS, well, and Eldre Creek waters. Concentrations and isotope ratios of well and Elder Creek samples are corrected for atmospheric input (see section S2 for formulation). Results of the vadose zone reactive transport model (solid) and the saturated zone batch-reactor model (dashed) are represented by black lines.

dust-derived input is a part of the local weathering cycle (Gaillardet et al., 1999). The Li concentrations ([Li]) are adjusted using a fraction of the dissolved chloride calculated from the molar Li/Cl ratio (4.6 x 10^{-5}) from the modern average seawater composition (Millero, 2016):

$$[\mathrm{Li}]^* = [\mathrm{Li}] - [\mathrm{Cl}] \left(\frac{[\mathrm{Li}]}{[\mathrm{Cl}]}\right)_{\mathrm{seawater}} \tag{S1}$$

Then, the δ^7Li values are corrected by assuming that the measured signature is a binary mixture of marine and silicate weathering inputs

$$\delta^7\mathrm{Li}_{\mathrm{measured}} = \delta^7\mathrm{Li}_{\mathrm{weathering}} f + \delta^7\mathrm{Li}_{\mathrm{seawater}}(1-f), \tag{S2}$$

where f is $\frac{[\mathrm{Li}]^*}{[\mathrm{Li}]}$ and $\delta^7\mathrm{Li}_{\mathrm{seawater}} = +31‰$ (Millot et al., 2004; Misra and Froelich, 2012).

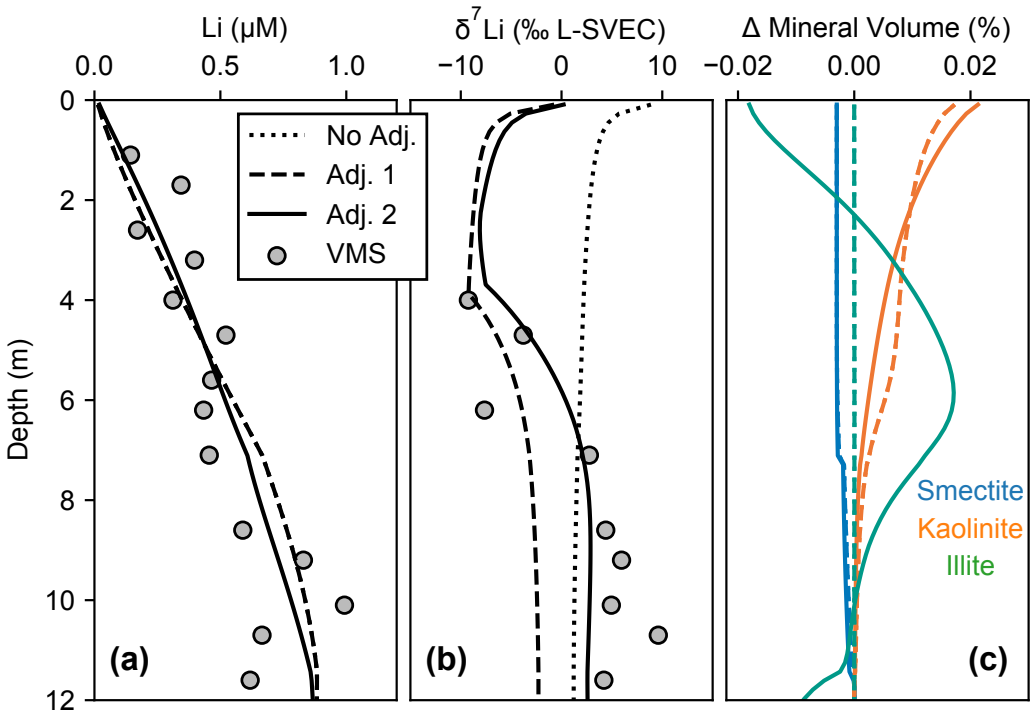

Figure S2: Observations (circles) and reactive transport model results (lines) of dissolved Li concentrations (a) and isotopic signatures (b) of VMS samples in the vadose zone. Panel (c) shows depth profiles of modeled changes in volume fraction for Li-bearing minerals over five complete pore volume flushes (subsubsection 5.1.1). Dotted profiles (No Adj.) denote the state of the model prior to any tuning against Li observations. Dashed profiles (Adj. 1) show the effect of adjusting the isotopic composition of the dissolving clays (i.e., smectite and illite) in the shallow portion of the domain from -0.57‰ to -12‰. The solid lines (Adj. 2) represent the final state of the model, including Adj. 1 plus an increase of the illite rate constant from $10^{-14.8}$ to $10^{-11.4}$ mol/m^2/s. See main text (subsection 4.2 and subsubsection 5.1.1) for a more detailed description.

To account for dilution, Li concentrations are normalized to those of sodium (Na). Such normalization also allows for characterization of the relative mobility of Li during silicate weathering, as Na is assumed to behave conservatively upon release via dissolution (Sawhney, 1972). Na concentrations are subjected to the same atmospheric input correction per Equation S1.

S3 Sensitivity analysis

S3.1 Fractionation factors

The resulting isotopic signature of the fluid (Figure 4b) is a balance of simultaneous dissolution and precipitation reactions. This balance is present throughout the entirety of the depth profile, even across the deeper portion of the domain (∼10-12 m), where it appears there is no formation of Li-bearing phases (Figure 4c). Although occurring minimally, the precipitation

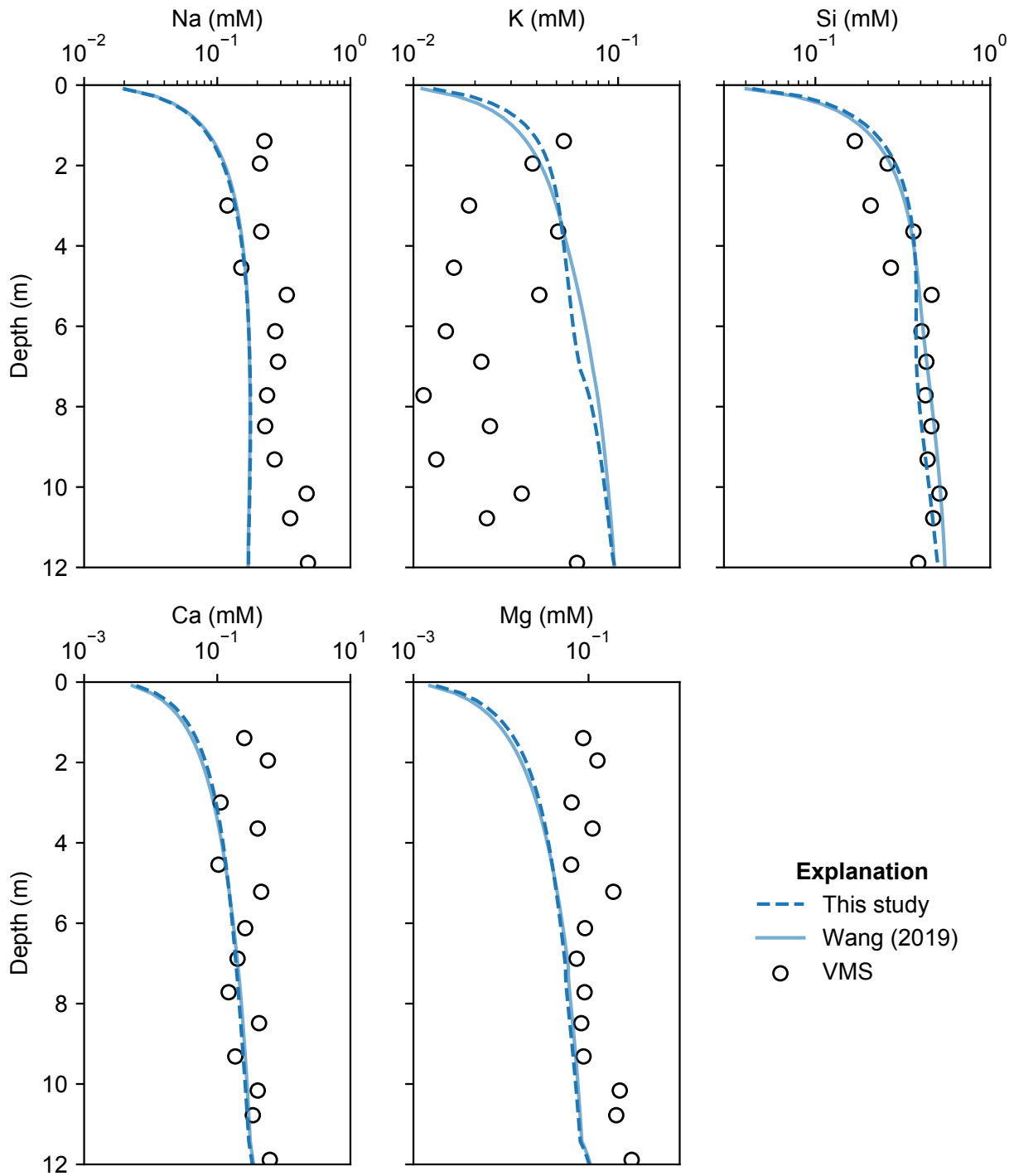

Figure S3: Observations (open circles) and reactive transport model results (lines) of vadose zone major solute concentrations. Each measured data point is an average of montly Vadose Zone Monitoring System samples from 2016 to 2017 from Wang (2019).

of kaolinite is competing against the dissolution of material with a fresh bedrock isotopic composition (i.e., illite and, to an extent, smectite). This isotopic fractionation associated with the formation of kaolinite is further demonstrated by Figure 4d. To demonstrate the significance of the simultaneous effects of dissolution and precipitation, the model is re-run with no isotopic fractionation during secondary mineral formation ($\alpha_{\text{sec-diss}} = 1$; Figure S4). The results of this simulation indicate that the resulting isotopic signature of the fluid is governed by the bulk isotopic composition of the dissolving phases, thus faithfully returning process 1 (Figure 2; section 1). With further dissolution across depth, the fluid composition approaches the fresh bedrock isotopic ratio. However, without any isotopic fractionation, the model is unable to reproduce the majority of the observed fluid isotope ratios and only captures the shallow negative values, which are interpreted to be a consequence of dissolution.

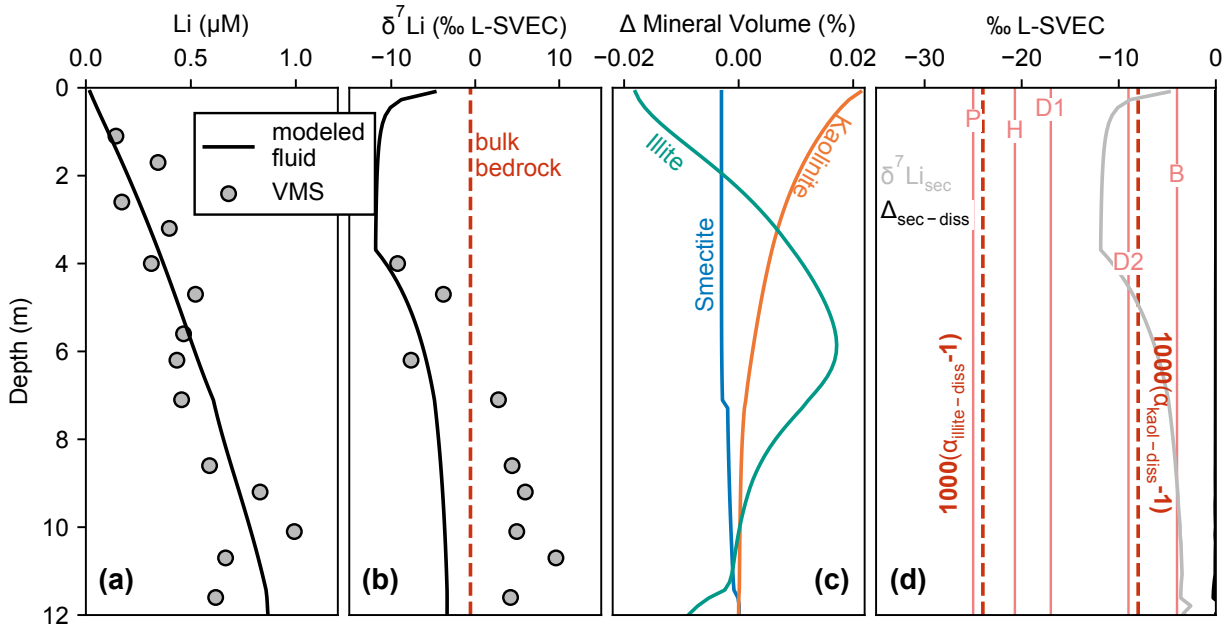

Figure S4: A version of the simulation where there is no isotopic fractionation during secondary mineral formation. Observations (circles) and reactive transport modeling results (solid lines) of dissolved Li concentrations (a) and isotopic signatures (b) of VMS samples in the vadose zone. In panel (b), the isotopic composition of the shale protolith is also marked as a red dashed line. Panel (c) shows simulated changes in mineral volume of Li-bearing phases with depth after five complete pore volume flushes (subsubsection 5.1.1). Panel (d) shows the estimated Li isotope composition for a new bulk secondary phase $\delta^7\text{Li}_{\text{sec}}$, and the 'effective' isotopic fractionation between the new bulk secondary phase and fluid, $\Delta_{\text{sec-diss}}$ (= $\delta^7\text{Li}_{\text{sec}}$ - $\delta^7\text{Li}_{\text{diss}}$; Eqn. 5). The dashed red vertical lines represent the fractionation factors for illite and kaolinite assigned to the model whereas the lighter red, solid vertical lines denote some of the $\Delta_{\text{sec-diss}}$ observed across different sites as described by Figure 4.

On the other hand, individually varying the fractionation factors for kaolinite and illite demonstrates the predominance of a single phase over a specific depth interval and thus the important contribution of mineralogy to the overall isotopic fractionation (Figure S5).

The greatest effects of varying $\alpha_{\text{kaol-diss}}$ are observed along the upper portion of the profile where kaolinite precipitation is prominent (Figure S5a). To an extent, the bottom portion of the profile is also affected by an adjustment to $\alpha_{\text{kaol-diss}}$ because the fluid isotope ratio at the depth at which illite begins to form is altered. Imposing the same sensitivity test to the illite fractionation factor yields a complementary result (Figure S5b). The significant changes to the overall shift in fluid δ^7Li from ∼3 m to depth correspond to the zone where illite is predominantly forming. Conversely, there is no influence on the simulation across the upper portion of the domain where illite is dissolving and thus causing no isotopic fractionation. The range of results generated by these simulations highlights the importance of constraining the specific reactions that induce Li isotopic fractionation and the use of appropriate accompanying fractionation factors.

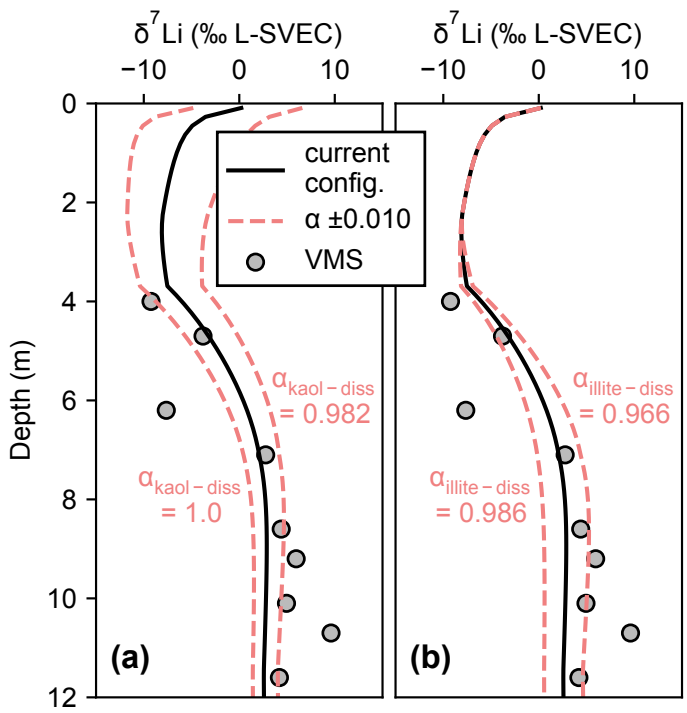

Figure S5: Results of the sensitivity analysis of individual fractionation factors. Observations (circles) and reactive transport modeling results (lines) of fluid isotopic signatures (b) of VMS samples in the vadose zone. The solid black line denotes the current working simulation whereas the dashed light red profiles refer to simulations where the fractionation factor is varied by ±0.010.

S3.2 Flow rate

We consider the effects of a variable flow rate on Li concentrations and isotope ratios by comparing the current working model to low- and high-flow cases (Figure S6). These simulations are run by setting the specific discharge ∼2x lower and higher relative to the value

in the main simulation (3 m/yr). The results illustrate that the concentration of Li in solution is a function of the contact time between the infiltrating fluid and undersaturated Li-bearing minerals (Figure S6a). If the fluid rapidly traverses the domain (high flow), a lower Li concentration is generated. The opposite case is true at low flow.

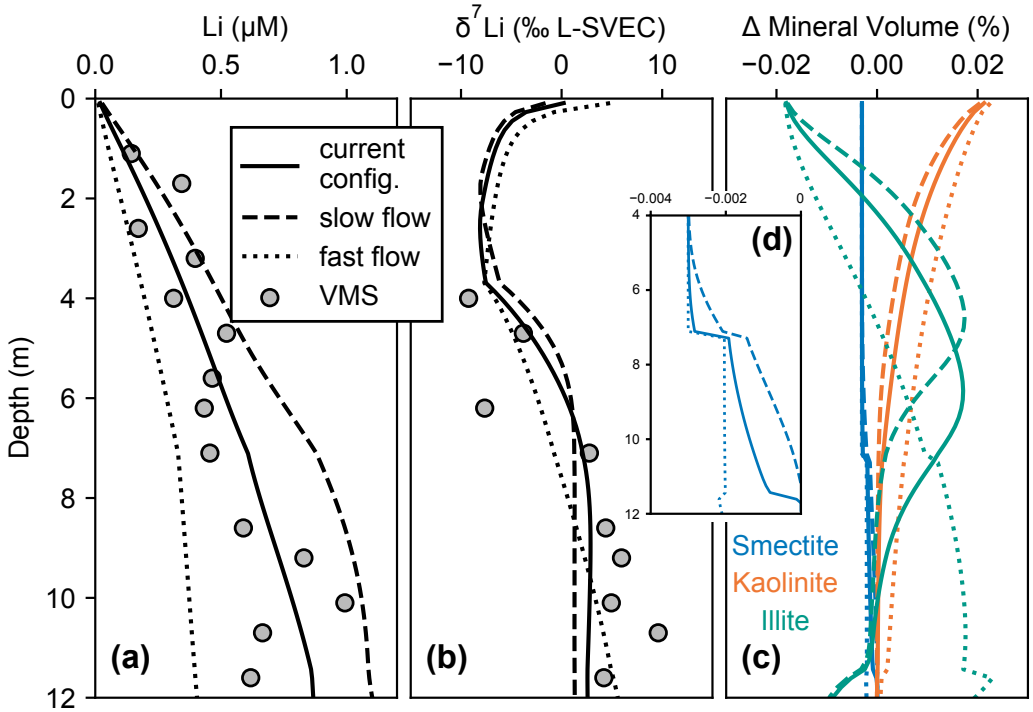

Figure S6: Results of the sensitivity analysis of flow rates. Observations (circles) and reactive transport modeling results (lines) of fluid isotopic signatures (b) of VMS samples in the vadose zone. Panel (c) shows simulated changes in mineral volume of Li-bearing phases with depth after five complete pore volume flushes (subsubsection 5.1.1). Panel (d) is an inset figure of panel (c) to highlight the changes in smectite mineral volume over the 4-12 m depth interval.

The influence of flow rate on the fluid δ^7Li is more complex (Figures S6b and S6c). Increasing fluid drainage rates pushes the illite reaction front deeper into the weathering profile and elongates the interval of depth over which kaolinite is formed. Whereas illite precipitation is associated with a Li fractionation factor of $\alpha_{\text{illite-diss}} = 0.976$, the value for kaolinite is $\alpha_{\text{kaol-diss}} = 0.992$. In tandem, the point at which the dissolution rate of smectite begins to decrease is also translated deeper with increasing flow rates (Figure S6d). Notably, this decrease becomes steeper but smoother at slower flow, mirroring the progressive decline of illite formation towards the bottom of the profile. This relationship highlights the duality of silicate weathering processes, given that dissolution and precipitation reactions modulate each other (Maher et al., 2009). In total, these coupled effects dictate the trend of fluid isotope ratios. For the high-flow simulation, a more gradual increase in δ^7Li beyond 4 m is

observed. In comparison, the resulting fluid profile from the low-flow case is not significantly different from that produced by the current working simulation. However, the shallower illite reaction front created at slower flow causes the positive shift in fluid δ^7Li to peak at a slightly shallower depth (∼6 m). Overall, these model tests highlight the sensitivity of Li isotopes to changes in the distribution of specific fluid-rock interactions during hydrologic transience in a multi-mineral system.

S4 Supplemental discussion of simulated mineral profiles

The presumed theoretical distribution of the secondary mineral weathering byproducts is linked to the aqueous chemistry of the infiltrating fluid, or, more specificially, to the activity of dissolved silica and the relative activity ratio of alkali species to the hydrogen ions in solution (Drever, 1988; Langmuir, 1997).

S4.1 Vadose zone

The first secondary mineral to form in the upper portion of the model domain is kaolinite (Figures 4c and S7). Theoretically, this mineral is favored in this shallow section of the profile due in part to the dilute composition of the infiltrating water and to the acidity supplied by elevated CO_2 that may be facilitating rapid (Langmuir, 1997) and preferential (Chou and Wollast, 1985) removal of alkali elements from feldspar, resulting in a remnant Al-Si structure foundational to kaolinite formation. Conversely, illite and smectite readily solubilize at the top of the model profile. Compositionally, these clays consist of low ionic potential or weakly bonded cations, like K^+ and Mg^{2+}, that are more easily removed from the crystalline structure (Langmuir, 1997). As the infiltrating water accumulates more dissolved alkali cations and silica through these modeled dissolution reactions, the predominant precipitation regime shifts to the formation of illite. The attenuated precipitation of kaolinite is coincident to this change in geochemical composition and to the cessation of microcline solubilization. The only source of K^+ for illite precipitation observed form the solid-phase samples (Gu et al., 2020) is smectite, which continues to dissolve with depth. This clay transformation reaction is consistent with expectation (Altaner and Ylagan, 1997).

S4.2 Saturated zone

The first clay reactions in the batch-reactor simulation (Figure S8) are the precipitation of smectite, and, eventually through the advancement of the saturation states of the minerals, the formation of illite. The dissolved Li that sustains the formation of these two clays during the initial stages of the batch-reactor simulation is sourced from the solubilization

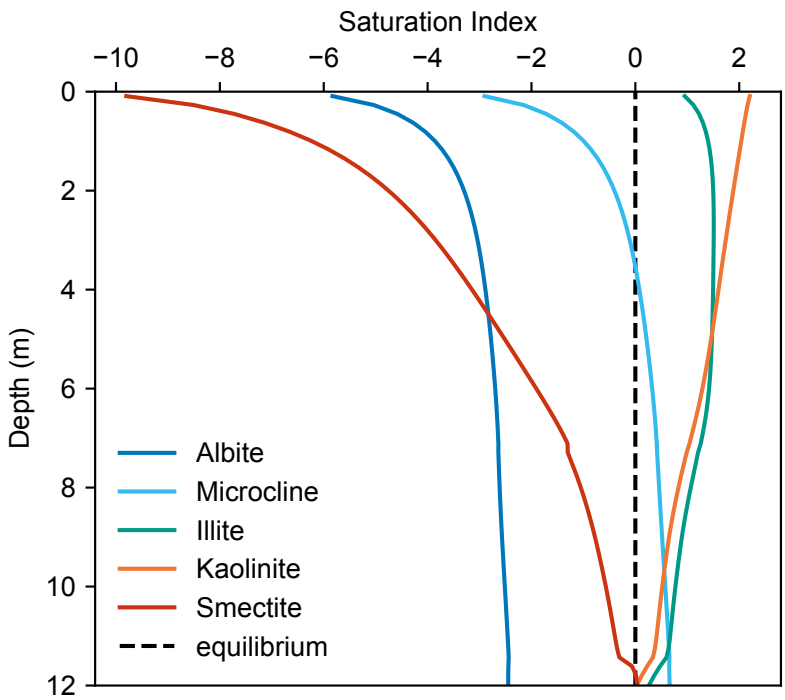

Figure S7: Saturation indices, or $\log(\mathrm{IAP}/\mathrm{K_{sp}})$, of Li-bearing secondary minerals and relevant primary phases calculated from modeled vaodse zone solute concentrations. The dashed black line denotes where $\log(\mathrm{IAP}/\mathrm{K_{sp}}) = 0$ or the state of thermodynamic equilibrium between the mineral and aqueous phases.

of kaolinite, which exists in the bedrock composition (Gu et al., 2020) rather than via prior formation in the vadose zone model. This dissolution is facilitated by an increase in pH likely due in part to the generation of carbonate alkalinity. The increasing pH and gradual enrichment of alkali solutes maintain the thermodynamic stability of the system away from kaolinite and towards clays like smectite and illite (Drever, 1988).

S5 Principal Component Analysis of Rivendell hydrogeochemistry

Principal Component Analysis or PCA (Hotelling, 1933; Jolliffe and Cadima, 2016) was used to summarize the variation of the overall hydrogeochemistry of the Rivendell waters to as few dimensions as possible. Available data from Table 2 are used in conjunction with major cation chemistry reported in Wang (2019) to constitute a continuous data matrix. Preceding PCA, these data are mean-centered and scaled to unit standard deviation. We apply the Kaiser-Guttman rule (Kaiser, 1960), which states that only principal components (PCs) with an eigenvalue or variance equal to or greater than 1 are retained, to arrive at two PCs that summarize ∼78% of the data structure (Figure S9). Variables (represented as gray lines) proximal to each other (i.e., Na, K, Ca, and Mg) are linearly correlated and vice-versa,

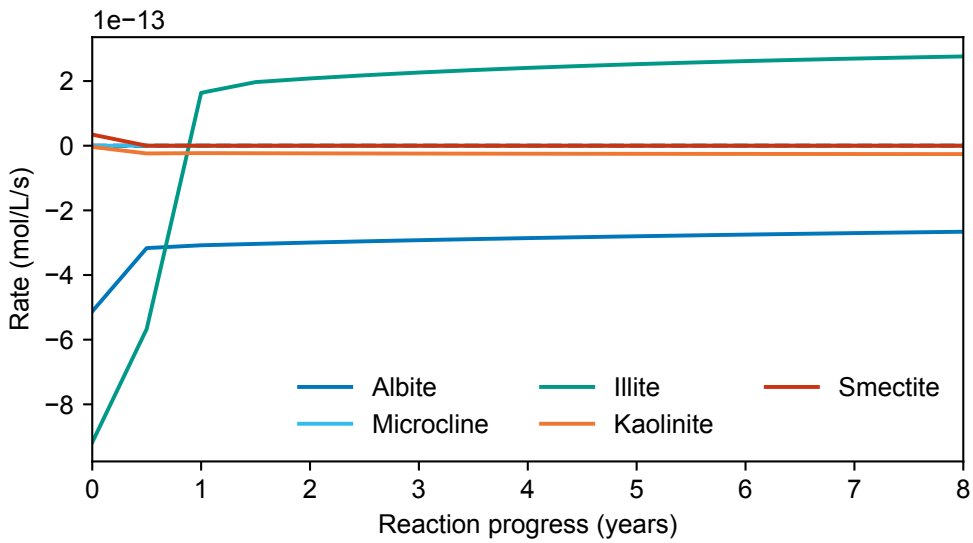

Figure S8: Modeled reaction rates of Li-bearing secondary minerals and relevant primary phases during progression of saturated zone batch-reactor model.

whereas samples (represented as colored markers) that trend along the direction of a given ray are associated to the corresponding variable (i.e., co-location of the samples taken from the saturated Vadose Zone Monitoring System ports and weathering-derived major ions).

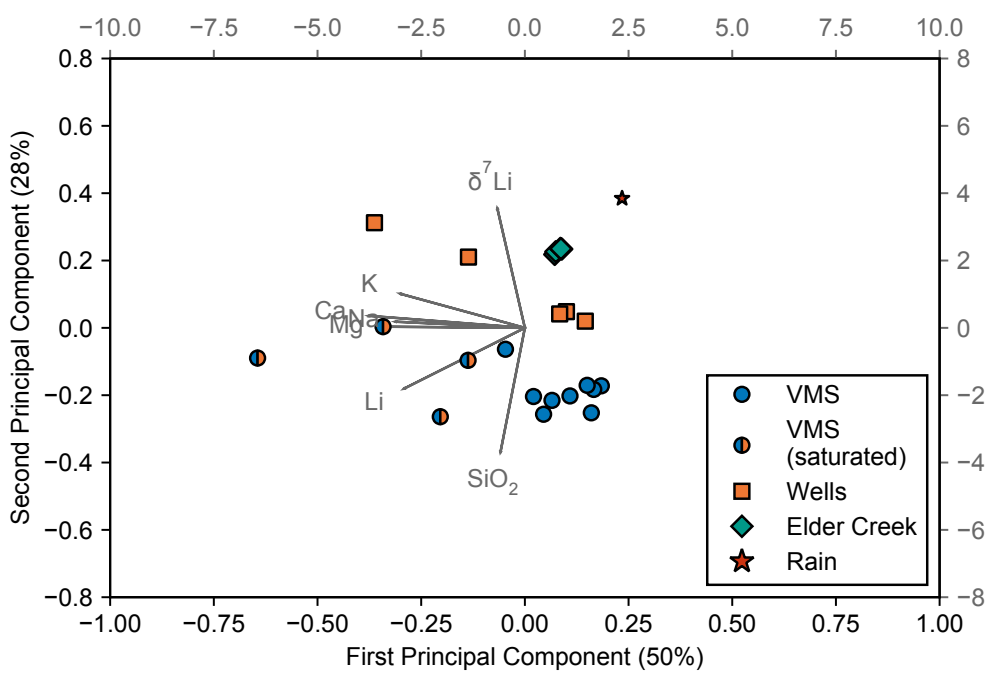

Figure S9: A biplot of the Rivendell hydrogeochemical data obtained from a Principal Component Analysis. The primary axes in black denote the projection of the samples (colored points) onto principal component space, whereas the secondary axes in gray represent the relative weightings of the geochemical variables (gray lines).

References

Altaner, S. P. and Ylagan, R. F. (1997). Comparison of Structural Models of Mixed-Layer Illite/Smectite and Reaction Mechanisms of Smectite Illitization. *Clays and Clay Minerals*, 45(4):517–533.

Chou, L. and Wollast, R. (1985). Steady-state kinetics and dissolution mechanisms of albite. *American Journal of Science*, 285(10):963–993. Publisher: American Journal of Science.

Dellinger, M., Gaillardet, J., Bouchez, J., Calmels, D., Louvat, P., Dosseto, A., Gorge, C., Alanoca, L., and Maurice, L. (2015). Riverine Li isotope fractionation in the Amazon River basin controlled by the weathering regimes. *Geochimica et Cosmochimica Acta*, 164:71–93.

Drever, J. I. (1988). *The Geochemistry of Natural Water*. Pearson Education Canada.

Fernandez, N. M., Zhang, X., and Druhan, J. L. (2019). Silicon isotopic re-equilibration during amorphous silica precipitation and implications for isotopic signatures in geochemical proxies. *Geochimica et Cosmochimica Acta*, 262:104–127.

Frings, P. J., Clymans, W., Fontorbe, G., De La Rocha, C. L., and Conley, D. J. (2016). The continental Si cycle and its impact on the ocean Si isotope budget. *Chemical Geology*, 425:12–36.

Gaillardet, J., Dupré, B., Louvat, P., and Allègre, C. J. (1999). Global silicate weathering and CO2 consumption rates deduced from the chemistry of large rivers. *Chemical Geology*, 159(1):3–30.

Gu, X., Rempe, D. M., Dietrich, W. E., West, A. J., Lin, T.-C., Jin, L., and Brantley, S. L. (2020). Chemical reactions, porosity, and microfracturing in shale during weathering: The effect of erosion rate. *Geochimica et Cosmochimica Acta*, 269:63–100.

Hindshaw, R. S., Tosca, R., Goût, T. L., Farnan, I., Tosca, N. J., and Tipper, E. T. (2019). Experimental constraints on Li isotope fractionation during clay formation. *Geochimica et Cosmochimica Acta*, 250:219–237.

Horstman, E. L. (1957). The distribution of lithium, rubidium and caesium in igneous and sedimentary rocks. *Geochimica et Cosmochimica Acta*, 12(1):1–28.

Hotelling, H. (1933). Analysis of a complex of statistical variables into principal components. *Journal of Educational Psychology*, 24(6):417–441. Place: US Publisher: Warwick & York.

Jolliffe, I. T. and Cadima, J. (2016). Principal component analysis: a review and recent developments. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 374(2065):20150202. Publisher: Royal Society.

Kaiser, H. F. (1960). The Application of Electronic Computers to Factor Analysis:. *Educational and Psychological Measurement*, 20(1):141–151. Publisher: Sage PublicationsSage CA: Thousand Oaks, CA.

Langmuir, D. (1997). *Aqueous Environmental Geochemistry*. Prentice Hall, NJ.

Li, W. and Liu, X.-M. (2020). Experimental investigation of lithium isotope fractionation during kaolinite adsorption: Implications for chemical weathering. *Geochimica et Cosmochimica Acta*, 284:156–172.

Maher, K., Steefel, C. I., White, A. F., and Stonestrom, D. A. (2009). The role of reaction affinity and secondary minerals in regulating chemical weathering rates at the Santa Cruz Soil Chronosequence, California. *Geochimica et Cosmochimica Acta*, 73(10):2804–2831.

Millero, F. J. (2016). *Chemical oceanography*. CRC press.

Millot, R. and Girard, J. (2007). Lithium isotope fractionation during adsorption onto mineral surfaces. In *International Meeting on Clays in Natural & Engineered Barriers for Radioactive Waste Confinement*, Lille, France.

Millot, R., Guerrot, C., and Vigier, N. (2004). Accurate and High-Precision Measurement of Lithium Isotopes in Two Reference Materials by MC-ICP-MS. *Geostandards and Geoanalytical Research*, 28(1):153–159.

Millot, R., Petelet-Giraud, E., Guerrot, C., and Négrel, P. (2010). Multi-isotopic composition (δ7Li–δ11B–δD–δ18O) of rainwaters in France: Origin and spatio-temporal characterization. *Applied Geochemistry*, 25(10):1510–1524.

Misra, S. and Froelich, P. N. (2012). Lithium Isotope History of Cenozoic Seawater: Changes in Silicate Weathering and Reverse Weathering. *Science*, 335(6070):818–823.

Odom, I. E., Fowden, L., Barrer, R. M., and Tinker, P. B. (1984). Smectite clay minerals: properties and uses. *Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences*, 311(1517):391–409. Publisher: Royal Society.

Pistiner, J. S. and Henderson, G. M. (2003). Lithium-isotope fractionation during continental weathering processes. *Earth and Planetary Science Letters*, 214(1):327–339.

Pogge von Strandmann, P. A. E., Renforth, P., West, A. J., Murphy, M. J., Luu, T.-H., and Henderson, G. M. (2021). The lithium and magnesium isotope signature of olivine dissolution in soil experiments. *Chemical Geology*, 560:120008.

Ronov, A., Migdisov, A., Voskresenskaya, N., and Korzina, G. (1970). Geochemistry of lithium in the sedimentary cycle. *Geochemistry International*, 7:75–102.

Sawhney, B. L. (1972). Selective Sorption and Fixation of Cations by Clay Minerals: A Review. *Clays and Clay Minerals*, 20(2):93–100.

Starkey, H. C. (1982). The role of clays in fixing lithium. Technical Report 1278-F, U.S. Geological Survey. Publication Title: Bulletin.

Vigier, N., Decarreau, A., Millot, R., Carignan, J., Petit, S., and France-Lanord, C. (2008). Quantifying Li isotope fractionation during smectite formation and implications for the Li cycle. *Geochimica et Cosmochimica Acta*, 72(3):780–792.

Wang, J. (2019). Linkages between fluid flow paths, reactive gases, and chemical weathering across a shale bedrock hillslope. Master's thesis, University of Illinois at Urbana-Champaign.

Williams, L. B. and Hervig, R. L. (2005). Lithium and boron isotopes in illite-smectite: The importance of crystal size. *Geochimica et Cosmochimica Acta*, 69(24):5705–5716.

Wimpenny, J., Colla, C. A., Yu, P., Yin, Q.-Z., Rustad, J. R., and Casey, W. H. (2015). Lithium isotope fractionation during uptake by gibbsite. *Geochimica et Cosmochimica Acta*, 168:133–150.

Zhang, L., Chan, L.-H., and Gieskes, J. M. (1998). Lithium isotope geochemistry of pore waters from ocean drilling program Sites 918 and 919, Irminger Basin. *Geochimica et Cosmochimica Acta*, 62(14):2437–2450.

Zhang, X. Y., Saldi, G. D., Schott, J., Bouchez, J., Kuessner, M., Montouillout, V., Henehan, M., and Gaillardet, J. (2021). Experimental constraints on Li isotope fractionation during the interaction between kaolinite and seawater. *Geochimica et Cosmochimica Acta*, 292:333–347.